Projection-Induced SVD and Boundary Metric: A Rank-Five Fibonacci-Autosimilar Characterization

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Abstract

The truncation rank of a singular value decomposition (SVD) is conventionally chosen by empirical thresholds or heuristic criteria. This paper does not establish a universal rank-selection law for compact operators. It characterizes a specific admissibility class of Fibonacci-autosimilar compact maps — finite-resolution linear maps representing the Fibonacci scale hierarchy $\{x_k = \varphi^k\}$ — and proves that the only SVD truncation rank compatible with autosimilar closure and metric stability within this class is $r = 5$.

The admissibility class is defined by three conditions on the map: orthogonal balance between retained and discarded spectral components (standard SVD truncation), the Fibonacci-autosimilar singular spectrum $\sigma_k = \varphi^{-k}$, and consistency of the induced metric with the spectral resolution. Within this class the admissible rank equals the cardinality of the α-resolvable transition spectrum: ranks below five exclude a transition that lies above the resolution threshold, while ranks above five retain modes that the projection cannot metrically distinguish.

Once $r = 5$ is established, the truncated projection P_5 canonically determines a logarithmic observable basis, a boundary metric $K = P_5^* P_5$ on the quotient space $\mathcal{H}/\ker P_5$, and a one-pole rational transfer filter $H(z) = P_4(z)/(1 - \alpha z)$: the column space, the quadratic form, and the scalar transfer function of the same operator.

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1 Introduction

The truncation rank of a singular value decomposition is one of the central decisions in finite-rank operator approximation. In practice it is an empirical choice: one fixes an energy threshold at 90% or 99%, applies an L-curve criterion [7], or looks for a visible gap in the spectrum. All of these approaches treat the rank as a free parameter to be tuned from outside the operator.

Finite-rank SVD for structured operators has been studied in related settings: Young [5] reduces the SVD of an infinite Hankel matrix of known rank r to an $r \times r$ problem; Macías-Virgós et al. [8] construct SVDs adapted to symplectic structure. In these settings, the rank is part of the input structure. The present paper identifies a class of compact operators for which the rank is instead forced by the operator's own algebraic and metric consistency: no external criterion is needed.

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The class consists of compact maps representing the Fibonacci-autosimilar scale hierarchy $\{x_k = \varphi^k\}$, where $\varphi = (1+\sqrt{5})/2$, under three structural constraints. Within this class, the unique admissible rank is $r = 5$. Both bounds come from the position of a single structural threshold, the inverse square of the Pincherle projection scale

$$\alpha = \Lambda^{-2}, \qquad \Lambda = 2\varphi^2\kappa,$$

where $2\varphi^2 = \mathcal{D}(\varphi^2)$ is the Pincherle scale increment ($\mathcal{D} = x\,d/dx$) and $\kappa = \sqrt{5}$ is the Fibonacci curvature; numerically $\alpha \approx 7.295 \times 10^{-3}$ (the closed form is derived in Section 2). The threshold acts within the spectrum of SVD transition weights $w_k = \sigma_k\sigma_{k+1} = \varphi^{-(2k+1)}$. The admissible rank is the cardinality of the resolvable transition spectrum, $r = \#\{k : w_k \geq \alpha\}$: the chain $w_4 > \alpha > w_5$ forces $r = 5$. Retaining fewer than five modes excludes a transition above threshold; retaining more assigns metric weight to configurations the projection cannot distinguish. Both violate the admissibility conditions. The selected rank coincides *a posteriori* with the discriminant $\Delta = |\lambda_+ - \lambda_-|^2 = 5$ of the Fibonacci recursion $v_{k+2} = v_{k+1} + v_k$, a closure capacity rather than a dimension; this agreement is recorded as a corollary and is not used to obtain the rank.

This is the main result of the paper (Theorem 4). Once $r = 5$ is fixed, three further objects are determined by the same truncated projection $P_5 = U_5\Sigma_5V_5^*$, without additional assumptions: the logarithmic observable basis $V_5 = \mathrm{span}\{(\log x)^n\}_{n=0}^4$, the boundary metric $K = P_5^*P_5$ on the quotient $\mathcal{H}/\ker P_5$, and the one-pole rational filter $H(z) = P_4(z)/(1-\alpha z)$. The rank selection, the induced metric, and the rational filter are three representations of a single finite-rank operator.

The Fibonacci-autosimilar class appears in systems where propagating and evanescent modes coexist with a scale hierarchy governed by the Fibonacci recursion. Two illustrative examples are given: acoustic radiation modes of a vibrating structure (Section 5.1), where the SVD of the radiation operator separates radiating from non-radiating velocity patterns, and Fibonacci photonic transfer spectra (Section 5.2), where the transfer hierarchy is modeled by the same φ^{-k} admissibility spectrum. In both cases the rank-5 truncation follows from the operator structure once the Fibonacci-autosimilar spectral condition holds.

Scope and independence. The analysis is operator-theoretic and model-independent. No physical carrier, gauge interpretation, dynamical equation, or field-theoretic ontology is used or assumed. The three admissibility requirements (I)–(III) are structural conditions on the SVD truncation of a compact map; they do not presuppose any specific physical realization of the φ-autosimilar hierarchy. The choice of $\varphi = (1+\sqrt{5})/2$ is not a physical assumption: it is the unique positive root of $\lambda^2 = \lambda + 1$, which is the minimal polynomial of a two-term self-referential recursion with scale-neutral representation (Section 2). Any physical or geometric system whose operator spectrum satisfies $\sigma_k \approx \varphi^{-k}$ inherits the results of Theorem 4 as a structural consequence, without requiring the derivation to be repeated.

The paper is organized as follows. Section 2 derives the non-isometric projection and the structural threshold α. Section 3 states the three admissibility requirements and proves $r = 5$. Section 4 derives the observable basis, boundary metric, and rational filter. Section 5 presents two illustrative applications: acoustic radiation modes (Section 5.1) and Fibonacci photonic transfer spectra (Section 5.2). Appendix A gives the numerical implementation. Appendix B derives the optional geometric readout (angular catalog and circular Vandermonde structure).

2 Primitive Closure and Non-Isometric Projection

Many scale-structured phenomena — from multiresolution signal decompositions to aperiodic optical systems — are governed by a hierarchy $\{x_k\}$ in which consecutive levels share a fixed ratio. When that ratio is the golden mean φ, the hierarchy is Fibonacci-autosimilar and the question is how to represent it by a finite-rank linear map between Hilbert spaces while controlling what is lost in the process. This section fixes φ, establishes the structure of such a map, and derives the threshold α that Section 3 uses to answer that question.

The goal is to represent the scale hierarchy $\{x_k = \varphi^k\}_{k\geq 0}$ by a linear map between Hilbert spaces. The ratio between consecutive levels is $x_{k+1}/x_k = \varphi$, so φ determines the entire hierarchy. Requiring that two consecutive levels determine the next without any additional parameter means $\varphi^2 = \varphi + 1$, which fixes φ uniquely as the positive root

$$\varphi = \frac{1+\sqrt{5}}{2}. \tag{1}$$

This equation has two roots $\lambda_+ = \varphi$ and $\lambda_- = -\varphi^{-1}$, and forces the scale levels to satisfy the Fibonacci recursion

$$v_{k+2} = v_{k+1} + v_k. \tag{2}$$

Their squared separation is

$$\Delta := (\lambda_+ - \lambda_-)^2 = 5. \tag{3}$$

Consider a compact linear operator $P : \mathcal{H} \to \mathcal{K}$ representing the φ-autosimilar scale hierarchy. Because φ is irrational, P cannot be isometric on the full hierarchy: any two φ-adapted steps $h = q_1\varphi^{n_1}$ and $\delta = q_2\varphi^{n_2}$ with $n_1 \neq n_2$ have irrational ratio, so no discrete grid can represent them exactly [4].

Treating all scale steps equally — no preferred level — requires the singular value of mode k to decay by the same factor φ^{-1} at each step, giving $\sigma_k = \varphi^{-k} \to 0$. This geometric decay is precisely the condition for P to be compact, and P admits the (infinite) singular value decomposition

$$P = U\,\Sigma\,V^*, \qquad \Sigma = \mathrm{diag}(\varphi^0, \varphi^{-1}, \varphi^{-2}, \ldots), \tag{4}$$

with orthonormal bases $\{v_k\}_{k\geq 0}$ in $\mathcal{H}$ and $\{u_k\}_{k\geq 0}$ in $\mathcal{K}$.

Rank-r truncation. Finite spectral resolution imposes a threshold $\sigma_{\rm cut} > 0$ below which singular modes are operationally indistinguishable. The effective representation is the *rank-r truncation*

$$P^{(r)} = U_r\,\Sigma_r\,V_r^*, \qquad \sigma_{r-1} \geq \sigma_{\rm cut} > \sigma_r, \tag{5}$$

where U_r, Σ_r, V_r retain the first r singular triples. The operator P is compact with infinitely many non-zero singular values; $P^{(r)}$ is its finite-rank approximation. Section 3 identifies the unique r compatible with the autosimilar structure. Elements of $\mathcal{H}$ that differ only within $\ker P^{(r)}$ are observationally indistinguishable at resolution r; the quotient $\mathcal{H}/\ker P^{(r)}$ records all operationally distinct configurations.

The generator of multiplicative rescaling on $x > 0$ is the operator

$$\mathcal{D} := x\,\frac{d}{dx} = \frac{d}{d(\log x)}, \tag{6}$$

which satisfies $\mathcal{D}(x^s) = s\,x^s$: monomials x^s are its eigenfunctions with eigenvalue s. In Section 4, restricting $\mathcal{D}$ to the r retained singular directions identifies the observable basis of P.

The two scale directions φ and φ^{-1} are reciprocals and together satisfy

$$\lambda^2 - \kappa\,\lambda + 1 = 0, \qquad \kappa = \varphi + \varphi^{-1}, \tag{7}$$

because their product is $\varphi \cdot \varphi^{-1} = 1$ and their sum is κ. From $\varphi - \varphi^{-1} = 1$ one gets $(\varphi + \varphi^{-1})^2 = (\varphi - \varphi^{-1})^2 + 4 = 5$, hence $\kappa = \sqrt{5}$. (This is a relation on the multipliers; the operator $\mathcal{D}$ is nilpotent on the polynomial basis and satisfies no such relation.)

The threshold α is a single number that separates, in Section 3, the singular modes P retains from those it discards. It is derived in three steps.

Step 1. The quadratic autosimilar level is represented by the monomial x^2. Applying $\mathcal{D}$ to this level and evaluating at $x = \varphi$ gives the elementary bulk scale increment:

$$\mathcal{D}(x^2)\big|_{x=\varphi} = 2\varphi^2. \tag{8}$$

Step 2. The spectral curvature of the Fibonacci recursion is $\kappa = \sqrt{5}$ (equation (3)). Using the identity

$$\sqrt{5} = 2\big(\varphi - \tfrac{1}{2}\big)\,, \tag{9}$$

this curvature is expressed directly in terms of the autosimilar scale variable φ. The bulk-to-boundary projection scale is fixed by the product of the Pincherle increment (8) and the Fibonacci spectral curvature:

$$\Lambda_{\mathrm{proj}} := \mathcal{D}(x^2)\big|_{x=\varphi} \cdot \kappa = 2\varphi^2\sqrt{5} = 4\varphi^2\big(\varphi - \tfrac{1}{2}\big)\,. \tag{10}$$

No numerical coefficient is introduced independently: Λ_{proj} is determined by the Pincherle dilation of the quadratic autosimilar level together with the spectral curvature.

Since φ satisfies $\varphi^2 = \varphi + 1$, the scale Λ_{proj} expands in $\mathbb{Q}(\sqrt{5})$ as

$$\Lambda_{\mathrm{proj}} = 4\varphi^2\big(\varphi - \tfrac{1}{2}\big) = 5 + 3\sqrt{5}. \tag{11}$$

With the standard Fibonacci convention ($F_0 = 0, F_1 = 1, F_2 = 1, \ldots$) one has $F_5 = 5$ and $F_4 = 3$, so $\Lambda_{\mathrm{proj}} = F_5 + F_4\sqrt{5}$. These Fibonacci coefficients are the $\mathbb{Q}(\sqrt{5})$-coordinates of the derived projection scale; they introduce no independent selection rule.

Step 3. The projection threshold is the inverse square of Λ_{proj}:

$$\alpha := \Lambda_{\mathrm{proj}}^{-2} = \big[4\varphi^2\big(\varphi - \tfrac{1}{2}\big)\big]^{-2}. \tag{12}$$

Since $\big(\varphi - \tfrac{1}{2}\big)^2 = \tfrac{5}{4}$, this reduces to

$$\alpha = \frac{1}{20\varphi^4} = \frac{7 - 3\sqrt{5}}{40} \approx 7.295 \times 10^{-3}. \tag{13}$$

(All three forms are identical; the numerical value follows from $\varphi^4 = 3\varphi + 2$ and $(7 - 3\sqrt{5})/40 = 1/(20\varphi^4)$, verified symbolically and numerically.) This number depends only on φ and lies in $\mathbb{Q}(\sqrt{5})$, the field of the Fibonacci recursion.

Remark 1 (Independence of α from the admissibility condition)**.** The threshold α is *not* defined by the condition $w_4 > \alpha > w_5$ that appears in the proof of Theorem 4. It is derived from two independent structural inputs:

(i) the elementary bulk scale increment $\mathcal{D}(x^2)\big|_{x=\varphi} = 2\varphi^2$, generated by the Pincherle operator at the quadratic autosimilar level φ^2 (Step 1);

(ii) the spectral curvature $\kappa = \sqrt{5}$, fixed by the discriminant $\Delta = (\lambda_+ - \lambda_-)^2 = 5$ of the Fibonacci recursion (equation (3)), independently of any rank or threshold.

Their product $\Lambda_{\text{proj}} = 4\varphi^2(\varphi - \frac{1}{2}) = 2\varphi^2\sqrt{5}$ is the unique bulk-to-boundary projection scale compatible with the autosimilar hierarchy and the Pincherle generator; $\alpha = \Lambda_{\text{proj}}^{-2} = [4\varphi^2(\varphi - \frac{1}{2})]^{-2}$ follows algebraically (Step 3).

The bilateral condition $w_4 > \alpha > w_5$ is then a *consequence*, not a definition. Numerically:

$$w_4 = \varphi^{-9} \approx 1.316 \times 10^{-2}, \qquad \alpha \approx 7.295 \times 10^{-3}, \qquad w_5 = \varphi^{-11} \approx 5.025 \times 10^{-3},$$

verified independently of the admissibility class. This independence is what makes Theorem 4 a genuine characterization result rather than a tautology.

Section 3 uses α to state the three admissibility requirements and proves that $r = 5$ is the unique rank compatible with all three.

3 Admissibility Class and Rank Selection

The threshold α derived in Section 2 now determines which truncations of the SVD are metrically consistent with the φ-autosimilar hierarchy.

The compact operator $P : \mathcal{H} \to \mathcal{K}$ admits the singular value decomposition

$$P = U\,\Sigma\,V^* \tag{14}$$

where:

- $V = [v_0\ v_1\ v_2\ \cdots]$ orthonormal right singular vectors in $\mathcal{H}$
- $U = [u_0\ u_1\ u_2\ \cdots]$ orthonormal left singular vectors in $\mathcal{K}$
- $\Sigma = \operatorname{diag}(\sigma_0, \sigma_1, \sigma_2, \ldots), \quad \sigma_0 \geq \sigma_1 \geq \cdots \geq 0$

$K := V\Sigma^2V^*$ has eigenvalues $\{\sigma_k^2\}$ and eigenvectors $\{v_k\}$. The concrete form of V, Σ, and $H(z)$ for the φ-autosimilar hierarchy is derived in Section 4.

The rank-r truncation is:

$$P^{(r)} = U_r\,\Sigma_r\,V_r^*, \qquad K^{(r)} = V_r\,\Sigma_r^2\,V_r^*. \tag{15}$$

Not every choice of r is meaningful. The following three requirements define the class of truncations for which rank and metric are mutually consistent. The rank is selected by Requirement (III) alone, as the cardinality of an α-resolvable transition spectrum; Requirements (I) and (II) supply the orthogonal split and the autosimilar singular spectrum on which (III) acts.

(I) Conservation balance. Let $P^{(r)} = U_r\Sigma_rV_r^*$ be the rank-r truncation of P. The SVD gives an orthogonal decomposition:

$$\|P^{(r)}f\|^2 + \|(P - P^{(r)})f\|^2 = \|Pf\|^2.$$

Equivalently, $\langle P^{(r)}f,\, (P - P^{(r)})f\rangle = 0$ for all $f \in \mathcal{H}$. This is the standard orthogonality of the retained and discarded singular components [2, 3]: the SVD truncation is the best rank-r approximation in operator norm (and in Hilbert–Schmidt norm when that norm is finite). This requirement does not select r; it supplies the orthogonal retained/discarded split on which Requirements (II) and (III) act.

(II) Fibonacci-autosimilar spectrum. The compact operator belongs to the φ-autosimilar class: its singular values follow the golden geometric law

$$\sigma_k = \varphi^{-k}, \qquad k = 0, 1, 2, \dots \tag{16}$$

or, at finite resolution, $\sigma_k^{(N)} = \varphi_N^{-k}$ with $\varphi_N = F_{N+1}/F_N$ the Fibonacci convergents and $\varphi = \lim_{N\to\infty} \varphi_N$ (Remark 2). This is the operator-theoretic content of the class: it fixes the singular spectrum and hence the transition weights w_k used in (III). It does not by itself select a rank. The scalar *closure capacity* $W := |\lambda_+ - \lambda_-|^2 = 5$ of the Fibonacci recursion (with branches $\lambda_+ = \varphi$, $\lambda_- = -\varphi^{-1}$) is a structural invariant of the class, not a dimension and not a count of singular directions; its agreement with the selected rank is recorded *a posteriori* in Corollary 5.

(III) Metric threshold stability. The boundary metric assigns diagonal weight σ_k^2 to mode k. The resolvability of the transition between adjacent scale levels $k \to k+1$ is measured by the mixed cell weight

$$w_k := \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)}. \tag{17}$$

The weight w_k is assigned to the *source level* k: it tests whether level k still carries a resolvable forward contrast before the next scale step is lost. It does not require level $k+1$ to be retained. Level k is metrically resolvable iff $w_k \geq \alpha$. Write the α-resolvable transition spectrum and the retained index set as

$$R_\alpha := \{k \geq 0 : w_k \geq \alpha\}, \qquad D_r := \{0, 1, \dots, r-1\}. \tag{18}$$

Metric stability is the single requirement that the retained modes coincide with the resolvable transition spectrum,

$$D_r = R_\alpha. \tag{19}$$

Equation (19) contains two clauses. Since w_k is strictly decreasing,

(III-a) *no over-resolution:* $k \in D_r \Rightarrow w_k \geq \alpha$, equivalently $w_{r-1} \geq \alpha$. Retaining a level whose forward weight lies below α makes the metric separate configurations the projection cannot resolve. This clause bounds r from above.

(III-b) *no under-resolution:* $w_k \geq \alpha \Rightarrow k \in D_r$, equivalently $w_r < \alpha$. Excluding a level whose forward weight lies above α collapses a genuine boundary distinction to zero. This clause bounds r from below.

Together: $w_{r-1} \geq \alpha > w_r$.

Remark 2 (Finite-resolution Fibonacci convergents)**.** The golden law (16) is the limit of a finite-resolution family. With $\varphi_N = F_{N+1}/F_N$ the Fibonacci convergents, $\sigma_k^{(N)} = \varphi_N^{-k}$ and $w_k^{(N)} = \varphi_N^{-(2k+1)}$, while the threshold takes the finite form $\alpha_N = [2\varphi_N^2(\varphi_N + \varphi_N^{-1})]^{-2} = F_N^6/[4F_{N+1}^2(F_{N+1}^2 + F_N^2)^2]$, with $\alpha = \lim_{N\to\infty} \alpha_N = 1/(20\varphi^4) = (7 - 3\sqrt{5})/40$. At the operationally natural level $N = 10$ ($F_{10} = 55$, $F_{11} = 89$, $\varphi_{10} = 89/55$) the threshold chain (20) is preserved exactly: $w_4^{(10)} \approx 1.3145 \times 10^{-2} > \alpha_{10} \approx 7.2916 \times 10^{-3} > w_5^{(10)} \approx 5.0200 \times 10^{-3}$, so the characterization $r = 5$ holds at finite resolution and in the limit alike. The closed form $\alpha = 1/(20\varphi^4)$ is a limit, not the primary definition; the radical form is recorded only for reference.

The transition weights $w_k = \varphi^{-(2k+1)}$ of (17) are computed from the singular values (16) alone. The threshold $\alpha = (7-3\sqrt{5})/40$ is derived in Section 2 from the Pincherle scale increment $2\varphi^2$ and the Fibonacci curvature $\kappa = \sqrt{5}$, independently of the admissibility class (Remark 1). Their comparison is therefore a numerical verification, not a circular definition:

$$w_4 = \varphi^{-9} \approx 1.316 \times 10^{-2} \;>\; \alpha \approx 7.295 \times 10^{-3} \;>\; \varphi^{-11} \approx 5.025 \times 10^{-3} = w_5. \tag{20}$$

Lemma 3 (Lower bound on truncation rank)**.** *Every finite-resolution representative $P^{(r)}$ of a map P in the admissibility class has truncation rank $r \geq 5$.*

Proof. The weights w_k are strictly decreasing in k, and $w_4 = \varphi^{-9} > \alpha$ by (20). By clause (III-b), every level with $w_k \geq \alpha$ must be retained; hence $k = 4 \in D_r$. Since $D_r = \{0, \ldots, r-1\}$, this forces $r - 1 \geq 4$, i.e. $r \geq 5$. The argument uses only the threshold position of α in the transition spectrum; it does not use the closure capacity W. □

Theorem 4 (Rank characterization within the admissibility class)**.** *Let $P : \mathcal{H} \to \mathcal{K}$ be a φ-autosimilar compact projection satisfying Requirements* (I)–(III)*. Then the retained index set coincides with the α-resolvable transition spectrum, and the unique admissible truncation rank is its cardinality,*

$$r = |R_\alpha| = \#\{k \geq 0 : \sigma_k \sigma_{k+1} \geq \alpha\} = 5. \tag{21}$$

This is a characterization result: it identifies the rank of every element of the admissibility class, not a universal rank-selection law for compact operators outside this class.

Proof. By (20) and the strict monotonicity of w_k, the superlevel set is exactly $R_\alpha = \{0, 1, 2, 3, 4\}$. Requirement (III), $D_r = R_\alpha$, then fixes $r = |R_\alpha| = 5$. Equivalently, in two bounds:

Lower bound ($r \geq 5$). $w_4 > \alpha$, so by clause (III-b) level $k = 4$ must be retained: $r \geq 5$ (Lemma 3).

Upper bound ($r \leq 5$). $w_5 < \alpha$, so by clause (III-a) level $k = 5$ must be excluded: were $r \geq 6$, the metric would assign positive weight to the sub-threshold transition w_5. Hence $r \leq 5$.

Conclusion. $r = 5$. □

Corollary 5 (Compatibility with the five-sector phasorial readout)**.** *The rank $r = 5$ selected by the metric threshold coincides with the Fibonacci closure capacity $W = |\lambda_+ - \lambda_-|^2 = 5$ of the recursion. The five retained modes therefore admit a phasorial realization in five sectors,*

$$\mathcal{V}_{\mathrm{ph}} = \bigoplus_{n \in \mathcal{N}_{\mathrm{adm}}} \mathbb{C}\, s_n, \qquad \mathcal{N}_{\mathrm{adm}} = \{2, 3, 5, 7, 11\}, \qquad \dim \mathcal{V}_{\mathrm{ph}} = 5,$$

detailed in Section B. This is a compatible reading of the already-selected rank, not a hypothesis used in the proof: the metric threshold fixes $r = 5$, and $r = 5$ in turn permits the five-sector closure. The reverse implication—closure capacity imposing the rank—is not asserted here; it is the route taken within the Majorana–Fibonacci–Heegner class of the companion rank-five note, on which the present characterization does not depend.

Theorem 4 fixes the rank but leaves open the question of what the operator P_5 concretely is. The answer is forced by the same algebraic constraints. A rank-5 operator $P_5 : \mathcal{H} \to \mathcal{K}$ has three canonical representations: its right singular subspace (a subspace of $\mathcal{H}$), the quadratic form $P_5^* P_5$ (a metric on the quotient $\mathcal{H}/\ker P_5$), and its scalar transfer function (a rational function on the autosimilar lattice). Each of these is uniquely determined by $r = 5$ and the Pincherle structure of Section 2: no additional assumption enters. Section 4 derives each in turn.

4 Observable Basis, Boundary Metric, and Rational Filter

With $r = 5$ fixed by Theorem 4 and the Pincherle operator $\mathcal{D} = d/d(\log x)$ from Section 2, the three canonical representations of P_5 are determined without further assumptions.

Observable basis.

Setting $y = \log x$ and $u = \log \varphi$, the autosimilar lattice becomes the arithmetic lattice $\{ku\}_{k \in \mathbb{Z}}$. The Pincherle operator acts as $\mathcal{D} = d/dy$ with Jordan ladder $\mathcal{D}(y^n) = n y^{n-1}$: since $\mathcal{D}$

lowers degree by one, any finite-dimensional downward-closed $\mathcal{D}$-invariant subspace containing constants has the form $\operatorname{span}\{1, y, \ldots, y^m\}$ for some m. The right singular vectors of P_5 span the $\mathcal{D}$-invariant subspace of P^*P; with $r = 5$, this is uniquely

$$V_5 = \operatorname{span}\{(\log x)^n\}_{n=0}^{4}, \tag{22}$$

the unique five-dimensional $\mathcal{D}$-invariant polynomial subspace containing constants consistent with $\{x_k = \varphi^k\}$. Modes $(\log x)^n$ with $n \geq 5$ lie in $\ker P_5$.

The logarithmic basis has a natural Mellin interpretation. Differentiating the Mellin character x^s with respect to its exponent gives

$$\frac{\partial^n}{\partial s^n} x^s = (\log x)^n \, x^s,$$

so the monomials $(\log x)^n$ are the polynomial jets of the Mellin character x^s. The retained modes $n = 0, \ldots, 4$ are precisely the five Mellin jets kept by the rank-five projection. The left singular vectors $U_5 = [u_0 \ \cdots \ u_4]$ are the images of V_5 under P_5 in $\mathcal{K}$; in the table below they are called Mellin-polynomial images because they carry the same finite-jet structure in the transform domain.

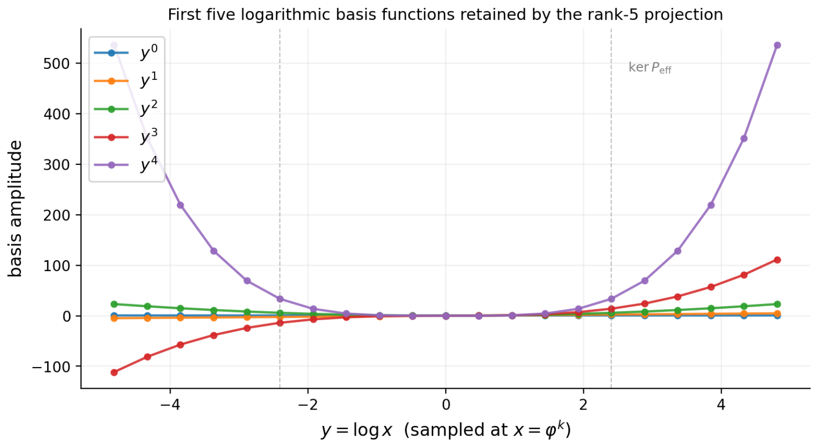

Figure 1: First five logarithmic basis functions retained by the rank-5 projection ($y = \log x$, sampled at $x = \varphi^k$). Modes $n \geq 5$ lie in $\ker P_5$.

Boundary metric.

The operator $K := P_5^* P_5 = V_5 \Sigma_5^2 V_5^*$ on $\mathcal{H}$, with $\sigma_i = \varphi^{-i}$, is positive semi-definite with $\ker K = \ker P_5$. On the quotient $\mathcal{H}/\ker P_5$ it is positive definite and defines the boundary metric

$$d_\partial([f],[g])^2 := (f-g)^* \, K \, (f-g) = \|P_5(f-g)\|_{\mathcal{K}}^2 = \|\Sigma_5 \, V_5^*(f-g)\|^2, \tag{23}$$

with $d_\partial([f],[g]) = 0$ if and only if $[f] = [g]$ in $\mathcal{H}/\ker P_5$. Under $x \mapsto cx$, y shifts by $\log c$; the eigenvalues $\{\sigma_i^2\}$ are invariant under such translation, so the spectral weights of d_∂ are scale-invariant on the retained autosimilar lattice. Two configurations are boundary-indistinguishable if and only if $f - g \in \ker P_5$. The vanishing of d_∂ is exact, not approximate. Components in $\ker P_5$ are not unresolved because of numerical precision or measurement noise; they lie outside the finite boundary representation. Refining the bulk description cannot give them positive boundary length unless the projection P_5 itself is changed. The measurable quantities at the boundary are represented by $\{(\log x)^n\}_{n=0}^4$ and the metric weights $\{\sigma_n^2\}_{n=0}^4$. The metric belongs to the class of scaled projections studied by Stewart [6]; it is the canonical member of that

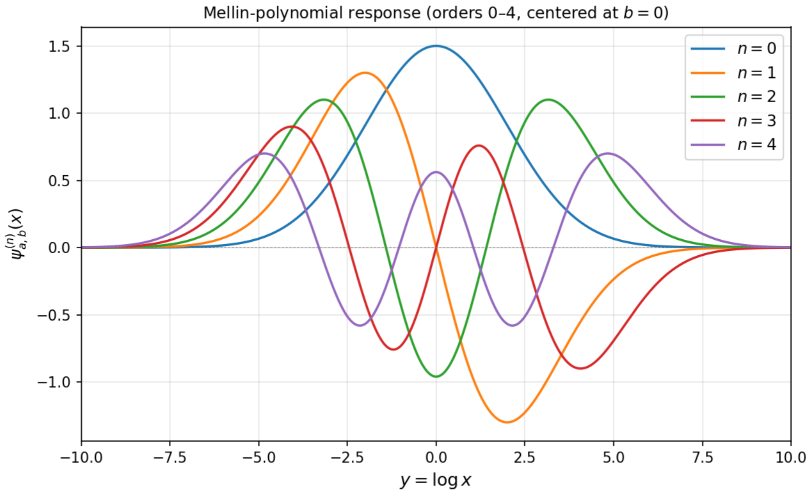

Figure 2: Mellin-polynomial response of the five retained logarithmic modes ($n = 0, \ldots, 4$, centered at $b = 0$).

class selected by the autosimilar admissibility conditions. The optional discrete boundary scales $\{\alpha \cdot 10^k\}_{k=0}^4$ are derived in Appendix B.

The threshold α and its operator lift.

The rank selection of Section 3 acts through a single scalar, the threshold α. It is convenient, and standard in DSP terms, to make explicit the operators on which α acts. Order the retained singular modes by the diagonal singular-value matrix

$$\Sigma = \operatorname{diag}(\sigma_0, \sigma_1, \sigma_2, \ldots), \qquad \sigma_k = \varphi^{-k}, \tag{24}$$

and let S be the one-step shift on the mode index, $S\, e_k = e_{k+1}$, i.e. the lower-shift matrix with $S_{k+1,k} = 1$ and zeros elsewhere. The *transition matrix* is the similarity-weighted shift

$$W := \Sigma\, S\, \Sigma, \qquad W_{k+1,k} = \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)} =: w_k, \tag{25}$$

a sub-diagonal (banded) matrix whose only non-zero entries are the adjacent-mode cell weights w_k. Explicitly, on the first six modes,

$$W = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \cdots \\ w_0 & 0 & 0 & 0 & 0 & \\ 0 & w_1 & 0 & 0 & 0 & \\ 0 & 0 & w_2 & 0 & 0 & \\ 0 & 0 & 0 & w_3 & 0 & \\ \vdots & & & & \ddots & \end{bmatrix}, \qquad \begin{aligned} w_0 &= \varphi^{-1}, & w_1 &= \varphi^{-3}, \\ w_2 &= \varphi^{-5}, & w_3 &= \varphi^{-7}, \\ w_4 &= \varphi^{-9}, & w_5 &= \varphi^{-11}. \end{aligned} \tag{26}$$

In this representation the rank is a *thresholded count* of the sub-diagonal: with $D_r = \{0, \ldots, r-1\}$ the retained index set and $R_\alpha = \{k : w_k \geq \alpha\}$ the α-superlevel set of W, Requirement (III) reads $D_r = R_\alpha$ and gives

$$r = \#\{k \geq 0 :\; W_{k+1,k} \geq \alpha\} = \#\{k :\; w_k \geq \alpha\} = 5, \tag{27}$$

the chain $w_4 > \alpha > w_5$ of (20) placing the cut between $k = 4$ and $k = 5$. The threshold acts on W exactly as a magnitude gate acts on the taps of a filter bank: taps with $|\cdot| \geq \alpha$ are kept, the rest are discarded.

Once the rank is fixed, the same threshold has fixed the truncated projection $P_5 = U_5\Sigma_5 V_5^*$ and hence the boundary metric as a matrix. In the orthonormal SVD basis the metric is diagonal,

$$K_{\mathrm{SVD}} = \Sigma_5^2 = \mathrm{diag}(1, \varphi^{-2}, \varphi^{-4}, \varphi^{-6}, \varphi^{-8}), \tag{28}$$

while in the (non-orthonormal) logarithmic working basis $b_i(y) = y^i$, $y = \log x$, $i = 0, \dots, 4$, it is the full Gram matrix

$$K_{ij} = \langle b_i, b_j\rangle_K = \langle P_5 b_i, P_5 b_j\rangle_{\mathcal{K}}, \qquad i, j = 0, \dots, 4, \tag{29}$$

a dense, symmetric positive-definite 5×5 matrix related to K_{SVD} by the change of basis B from $\{b_i\}$ to $\{v_i\}$, $K = B^* K_{\mathrm{SVD}}\, B$ (Hermitian conjugate, the working spaces being complex). Equations (28) and (29) are the same quadratic form in two coordinate systems: diagonal in the singular basis, full in the logarithmic basis a practitioner would actually sample.

Remark 6 (Channel permutations are not metric isometries)**.** The retained index set $\{0, 1, 2, 3, 4\}$ carries a natural action of the symmetric group Sym(5) permuting the five observable channels. This combinatorial symmetry is, however, broken by the metric: since the diagonal weights $\sigma_k^2 = \varphi^{-2k} \in \{1, \varphi^{-2}, \varphi^{-4}, \varphi^{-6}, \varphi^{-8}\}$ are pairwise distinct, the only permutation P_π with $P_\pi^* K_{\mathrm{SVD}} P_\pi = K_{\mathrm{SVD}}$ is the identity; the isometry subgroup of K within the channel set is trivial. The five channels are thus combinatorially equivalent but metrically ordered: the rank characterization fixes *which* five modes survive, while K fixes their weighted hierarchy. Any equal-weight (permutation-symmetric) reading of the five channels is a statement about the support of the projection, not about its boundary metric.

The three objects are therefore one structure in three matrix forms, selected by the single scalar α:

$$\underbrace{\alpha}_{\text{scalar threshold / pole}} \quad\longrightarrow\quad \underbrace{W = \Sigma S \Sigma}_{\text{transition matrix (gating)}} \quad\longrightarrow\quad \underbrace{K = P_5^* P_5}_{\text{boundary metric (Gram)}} \quad\longrightarrow\quad \underbrace{H(z)}_{\text{scalar transfer (pole/zero)}}. \tag{30}$$

The threshold α is scalar only as a gate level and as the pole location of $H(z)$; its action on the boundary is matrix-valued through W and through the metric Gram matrix K. It is K, not α, that is the boundary metric; α selects the projection that generates K.

Rational filter.

The hierarchy $\{x_k = \varphi^k\}$ is a multiplicative lattice. Setting $z_k := \varphi^{-k}$ maps it to a geometric sequence in $[0, 1]$. On this lattice, the kernel K evaluated at the base point $x_1 = \varphi$ generates the polynomial

$$P_4(z) := \sum_{n=0}^{4} (\log \varphi)^n\, z^n, \tag{31}$$

where each coefficient $(\log \varphi)^n$ is the n-th power of the additive lattice step $u = \log \varphi$. The projection has a single structural invariant $\Lambda_{\mathrm{proj}} = 5 + 3\sqrt{5}$, hence a single projection scale $\alpha = \Lambda_{\mathrm{proj}}^{-2}$ and a single pole at $z_{\mathrm{pole}} = \alpha^{-1}$. With the normalization $H(0) = 1$, the corresponding one-pole rational transfer representative is

$$H(z) = \frac{P_4(z)}{1 - \alpha z}, \qquad \alpha^{-1} = \Lambda_{\mathrm{proj}}^2 = (F_5 + F_4\sqrt{5})^2 = 70 + 30\sqrt{5}, \tag{32}$$

where $F_5 = 5$ and $F_4 = 3$ are adjacent Fibonacci numbers and $\Lambda_{\mathrm{proj}} = F_5 + F_4\sqrt{5} = 5 + 3\sqrt{5}$ is the structural invariant of the projection (10).

Proposition 7 (Zero–pole structure of the boundary filter)**.** *For the rank-five projection of Theorem 4, the boundary filter* $H(z) = P_4(z)/(1-\alpha z)$ *of* (32) *has the structure below, fixed entirely by the projection and requiring no rank- or realization-selection criterion external to it:*

(i) *a finite-order zero,* $\deg P_4 = r - 1 = 4$*, equivalent to the rank:* $\deg P_4 = 4 \Leftrightarrow \text{rank} = 5$*;*

(ii) *a single pole, as a count — the rational image of the one inversion-invariant scale carried by the projection;*

(iii) *no canonical scalar pole location: one and the same projection scale is read, equivalently, as the reciprocal golden branches* φ^{-1} *and* φ^2*, as* $\alpha^{-1} = \Lambda_{\text{proj}}^2$*, and on the boundary ladder* $\alpha \cdot 10^k$*,* $k = 0, \dots, 4$*.*

Proof. (i) Finite-order zero. The retained observable space $V_5 = \text{span}\{(\log x)^n\}_{n=0}^4$ has dimension $r = 5$. On the lattice the numerator carries its five coefficients $\{(\log\varphi)^n\}_{n=0}^4$ in the monomial frame $\{z^n\}$, so $\deg P_4 = r - 1 = 4$; a lower degree would drop a retained Mellin jet $(\log x)^n$, $n \le 4$, contradicting the rank-five identification of V_5. The numerator is thus a finite-order zero with $\deg P_4 = 4 \Leftrightarrow \text{rank} = 5$.

(ii) One pole, by count. The projection carries exactly one inversion-invariant scale. The two φ-directions form the reciprocal pair (φ, φ^{-1}) with product 1; under loss of transport orientation the inversion $z \mapsto z^{-1}$ exchanges the two branches, so the individual values φ, φ^{-1} are not retained, while their inversion-symmetric trace $\kappa = \varphi + \varphi^{-1} = \sqrt{5}$ (7) is. Pincherle dilation on the quadratic autosimilar level fixes $\mathcal{D}(x^2)|_{x=\varphi} = 2\varphi^2$ (8), and the unique bulk–boundary scale is its product with that surviving invariant, $\Lambda_{\text{proj}} = 2\varphi^2\sqrt{5} = 5 + 3\sqrt{5}$ (10), hence the single projection constant $\alpha = \Lambda_{\text{proj}}^{-2}$ (12). The numerator P_4 alone is a polynomial — a finite-memory transfer with no pole — transmitting the five retained modes but encoding no gate; a polynomial cannot represent the single contraction by which the projection separates retained from discarded modes, so any proper rational representative must carry at least one pole. The denominator records the projection's scales as its roots; with a single scale it is the single factor $1 - \alpha z$, and the pole count is one — necessary, to leave the polynomial class, and sufficient, since no second independent scale exists. This is a structural count, not a realization choice: it needs no minimal-realization, stability, or causality argument beyond the single-scale content already fixed.

(iii) No canonical scalar. Because the orientation information distinguishing φ from φ^{-1} is not retained, the scalar *value* at which the single pole is recorded is representation-dependent. The same projection scale α is assembled from the reciprocal golden branches $c_- = \varphi - 1 = \varphi^{-1}$ (contraction) and $c_+ = \varphi + 1 = \varphi^2$ (expansion) through $\alpha = \frac{1}{20} c_-^2/c_+ = \varphi^{-4}/20$ (39),(40); in the transfer form (32) the pole sits at $z = \alpha^{-1} = \Lambda_{\text{proj}}^2 = 70 + 30\sqrt{5}$; on the boundary-scale ladder it sits at $\alpha \cdot 10^k$ (Table 1). These are three readings of one relational invariant, not three poles. The invariant content of the filter is the inversion-symmetric scale Λ_{proj} together with the rank-five zero; the particular scalar at which the pole is recorded is a choice of representation. □

Remark 8 (Relational reading: invariant ratios, dispersed realizations)**.** Proposition 7(iii) has a direct metrological counterpart. A finite-rank projection of an irrational scale invariant fixes every dimensionless ratio of the retained spectrum — the singular ratios $\sigma_k/\sigma_{k+1} = \varphi$, the metric weights σ_k^2, the invariant scale Λ_{proj} — but does not fix an absolute linear unit: the invariant resides in the resolvable singular subspace, while the admissible absolute realizations are distributed within a bounded set whose spread is governed by the discarded mass M_{disc} (34). A single canonical unit cannot be recovered; what is stable is the relation, not the magnitude. This is the same fact as the representational multiplicity of the pole: the scalar at which the pole is recorded, and the absolute unit against which lengths are measured, are both representation-dependent, whereas the inversion-symmetric ratio content is invariant. Dispersion of absolute realizations is therefore not measurement error but the structural signature of finite-rank projection of an irrational invariant.

On the five-node lattice $\{x_k = \varphi^k\}_{k=0}^4$, let $\mathbf{\Delta f} = [f(x_0) - g(x_0),\ \ldots,\ f(x_4) - g(x_4)]^T$. The boundary metric is:

$$d_\partial(f,g)^2 = \mathbf{\Delta f}^* \, \mathbf{K} \, \mathbf{\Delta f}, \qquad K = V_5 \, \Sigma_5^2 \, V_5^*, \tag{33}$$

where $\Sigma_5 = \operatorname{diag}(\varphi^0, \varphi^{-1}, \varphi^{-2}, \varphi^{-3}, \varphi^{-4})$ and $V_5 = [v_0 \ \cdots \ v_4]$ with v_n obtained by orthonormalizing $\{(\log x)^n\}_{n=0}^4$ on the retained lattice.

The complement of the retained metric is recorded by a single basis-independent quantity. The modes beyond the fifth carry the *discarded spectral mass*

$$M_{\text{disc}} := \sum_{k \ge r} \sigma_k^2 = \sum_{k \ge 5} \varphi^{-2k} = \frac{\varphi^{-10}}{1 - \varphi^{-2}} = \varphi^{-9}, \tag{34}$$

using $1 - \varphi^{-2} = \varphi^{-1}$. This depends only on the spectrum of P^*P and is therefore coordinate-free; it measures the information suppressed by truncation, not a thermodynamic entropy. Its value coincides with the first sub-threshold transition weight, $M_{\text{disc}} = w_4 = \varphi^{-9}$, so the mass that the rank-five projection discards equals the weight of the cell it declines to resolve. The total spectral mass is $\sum_{k \ge 0} \varphi^{-2k} = \varphi$, whence the discarded fraction is $M_{\text{disc}}/\varphi = \varphi^{-10} \approx 0.81\%$: finite rank suppresses a small, exactly quantified remainder, not an arbitrary tail.

Let

$$\mathbf{p} = \begin{bmatrix} (\log \varphi)^0 \\ (\log \varphi)^1 \\ \vdots \\ (\log \varphi)^4 \end{bmatrix}, \qquad \mathbf{z}_k = \begin{bmatrix} 1 \\ z_k \\ z_k^2 \\ z_k^3 \\ z_k^4 \end{bmatrix}.$$

Then $P_4(z_k) = \mathbf{p}^T \mathbf{z}_k$, and the filter value at the lattice point $z_k = \varphi^{-k}$ is

$$H(z_k) = \frac{\mathbf{p}^T \mathbf{z}_k}{1 - \alpha z_k}. \tag{35}$$

The scalar action on a lattice component is

$$\Delta f_k \ \mapsto \ \frac{p^T z_k}{1 - \alpha z_k} \, \Delta f_k. \tag{36}$$

$H(z)$ is a *scalar transfer representative* of the finite-rank metric data of $K = P_5^* P_5$ on the autosimilar lattice: it encodes the same singular values and logarithmic basis as K, but as a rational function on $\{z_k = \varphi^{-k}\}$ rather than as an operator on $\mathcal{H}$. The representation is exact on the five-node lattice and does not claim a full operator identity between $H(z)$ and K. More precisely: K is a positive semi-definite operator on $\mathcal{H}$ with eigenvalues $\{\sigma_k^2 = \varphi^{-2k}\}$; when restricted to the five-node lattice $\{z_k = \varphi^{-k}\}_{k=0}^4$, the scalar weight it assigns to node k is $\sigma_k^2 = \varphi^{-2k}$. The filter $H(z_k) = P_4(z_k)/(1 - \alpha z_k)$ encodes these weights through its numerator — the polynomial P_4 evaluated at z_k gives the k-th basis coefficient — and its pole at $z = \alpha^{-1}$ controls the decay rate across the lattice. The two objects K and $H(z)$ are thus representations of the same finite-rank metric structure: K in operator form on $\mathcal{H}$, and $H(z)$ in scalar transfer form on the lattice $\{z_k\}$.

In the analytic (z) convention the pole of $H(z)$ is at $z = \alpha^{-1} = (F_5 + F_4\sqrt{5})^2 = 70 + 30\sqrt{5}$, outside the unit disk, so $H(z)$ is holomorphic on the closed unit disk $|z| \le 1$. This is analyticity, not causal stability: causal BIBO-stability is a property of the z^{-1} (causal DSP) convention, in which the pole sits at $z = \alpha$ strictly inside the disk (eq. (37) below). The two statements are not in tension — they are the same single pole read in reciprocal conventions.

In standard DSP form, the causal implementation reads the same transfer function in z^{-1},

$$H_c(z) = \frac{P_4(z^{-1})}{1-\alpha\, z^{-1}} = \frac{\sum_{n=0}^{4}(\log\varphi)^n z^{-n}}{1-\alpha z^{-1}}, \tag{37}$$

a degree-four FIR numerator (the five retained log-modes as taps) in cascade with a single one-pole IIR section. The causal pole sits at $z = \alpha \approx 7.29 \times 10^{-3}$, well inside the unit circle, so the causal filter is minimum-phase and BIBO-stable; the analytic (anti-causal) form (32) places the reciprocal pole $z = \alpha^{-1} \approx 137$ outside the disk, which is the property used for the boundary relation below.

Written as a recurrence on the scale index k, with input u_k, output y_k, and FIR taps $p_n = (\log\varphi)^n$, the causal filter reads

$$y_k \;=\; \sum_{n=0}^{4} p_n\, u_{k-n} \;+\; \alpha\, y_{k-1}, \tag{38}$$

the explicit form of the structure rank-five finite memory + one-pole memory: the sum is the finite (FIR) record of the five retained log-modes, and the single feedback term $\alpha\, y_{k-1}$ is the one-pole (IIR) memory carrying the projection scale α. No higher-order feedback appears, consistent with the single-pole count of Proposition 7(ii).

The threshold admits a closed golden form built from the reciprocal branches of the autosimilar scale, the *golden contraction/expansion pair*

$$c_- = \varphi - 1 = \varphi^{-1} \quad \text{(contraction)}, \qquad c_+ = \varphi + 1 = \varphi^2 \quad \text{(expansion)}, \tag{39}$$

using $\varphi^2 = \varphi + 1$, with three-step golden compression $c_-^2/c_+ = \varphi^{-4}$. These are not the zeros and poles of $H(z)$: the only pole of $H(z)$ is the reciprocal threshold $z = \alpha^{-1}$, and the zeros of H are the roots of the numerator P_4. The pair (c_-, c_+) supplies the structural normalization of the pole threshold α, namely the golden ratio scaled by the projection constant Λ^2 (the factor 20 in the limit, the same 20 as $\alpha = 1/(20\varphi^4)$ of Section 2),

$$\alpha = \frac{1}{20}\,\frac{c_-^2}{c_+} = \frac{\varphi^{-4}}{20} = \frac{1}{20\varphi^4} = \frac{7-3\sqrt{5}}{40}. \tag{40}$$

The normalizing $20 = 4\cdot 5$ is structural (4 from the Pincherle increment $2\varphi^2$ squared in Λ, 5 from the Fibonacci curvature $\kappa^2 = 5$); it is not a decimal coincidence and does not factor as $2\cdot 10$ with an independent decade. The golden branches $c_\pm$, the transfer pole α^{-1}, and the boundary ladder $\alpha \cdot 10^k$ are the three loci catalogued in Proposition 7(iii): not distinct poles, but representations of the one inversion-symmetric projection scale Λ, recorded at different points of the structure.

Analytic extension and Hilbert boundary relation. Since P_4 has real coefficients and the pole $z = \alpha^{-1}$ lies outside the unit disk, $H(z)$ is holomorphic in $|z| < 1$. Its boundary values on the unit circle therefore satisfy the circular Hilbert relation

$$\operatorname{Im} H(e^{i\omega}) = \mathcal{H}_{\mathrm{circ}}\big[\operatorname{Re} H(e^{i\omega})\big], \tag{41}$$

up to the constant fixed by $H(0) = 1$. The imaginary component is not an additional datum: it is determined by the real component through analyticity.

Remark 9 (Scale contrast at $z = 1$)**.** At $z = 1$ the one-pole factor reduces to $(1-\alpha)^{-1}$. The ratio $\rho := (\varphi-1)/(\varphi+1) = \varphi^{-3}$, where $\varphi-1 = \varphi^{-1}$ and $\varphi+1 = \varphi^2$ follow from $\varphi^2 = \varphi+1$, is the natural scale contrast between the reciprocal branch and the forward branch at the boundary point $z = 1$. It may be read as the fraction of the autosimilar hierarchy discarded by the rank-five truncation. This interpretation is not used in the proof of Theorem 4; it records a boundary reading of the scale split encoded by the rational filter.

In the present setting, $\mathrm{Re}\, H$ is the scalar transfer representative of the quotient metric $K = P_5^* P_5$, while $\mathrm{Im}\, H$ is the harmonic-conjugate trace of the discarded projection residual. This observation is not used in the rank-selection proof; it records that the rational representative of the finite-rank metric carries the standard analytic-signal structure of Hardy boundary values.

Summary: the complete structure $P = U_5\, \Sigma_5\, V_5^*$.

Object	Concrete form	Role
$V_5 = [v_0\ \cdots\ v_4]$	$v_n = \mathrm{orth}\{(\log x)^n\}$	right singular vectors
Σ_5	$\mathrm{diag}(\varphi^0, \varphi^{-1}, \varphi^{-2}, \varphi^{-3}, \varphi^{-4})$	singular values
$U_5 = [u_0\ \cdots\ u_4]$	Mellin-polynomial images in $\mathcal{K}$	left singular vectors
$W = \Sigma S \Sigma$	sub-diagonal, $W_{k+1,k} = \varphi^{-(2k+1)}$	transition matrix (α-gated)
$K = V_5 \Sigma_5^2 V_5^*$	boundary metric kernel	Theorem 4
K_{SVD}	$\mathrm{diag}(1, \varphi^{-2}, \varphi^{-4}, \varphi^{-6}, \varphi^{-8})$	metric, singular basis
K_{ij}	$\langle P_5 b_i, P_5 b_j \rangle$, dense 5×5	metric Gram, log basis
$d_\partial^2(f, g)$	$\Delta f^*\, K\, \Delta f$	boundary metric
p	$[1,\ \log\varphi,\ (\log\varphi)^2,\ (\log\varphi)^3,\ (\log\varphi)^4]^T$	numerator coefficients
$H(z_k)$	$p^T z_k\, /\, (1 - \alpha z_k)$	filter at $z_k = \varphi^{-k}$

The three representations above complete the algebraic characterization of P_5. Section B derives a further geometric consequence: the rank-five structure admits, under the optional angular readout of Appendix B, a finite catalog of admissible angular denominators on S^1, which connects the SVD to cyclic graph theory and circular Vandermonde matrices.

5 Illustrative Applications

The admissibility class characterized in Sections 3–B is illustrated by two physical systems in which a Fibonacci-autosimilar spectral condition of the form $\sigma_k \approx \varphi^{-k}$ can arise from the structure of the underlying operator. Both applications are conditional on that spectral condition; neither is a proof that the condition holds in full generality for these systems.

5.1 Acoustic radiation modes

The power radiated by a vibrating surface into an acoustic fluid is

$$W = \tfrac{1}{2}\mathbf{v}^H \mathbf{R}\, \mathbf{v}, \tag{42}$$

where $\mathbf{v}$ is the discretised normal surface velocity and $\mathbf{R}$ is the radiation resistance matrix. Borgiotti [11] showed that the SVD of the radiation operator yields orthonormal velocity patterns, the *Acoustic Radiation Modes* (ARMs), each contributing independently to the radiated power, with associated radiation efficiencies $\{\sigma_k^2\}$ ordered $\sigma_1^2 \geq \sigma_2^2 \geq \cdots \geq 0$. Photiadis [12] identified these singular values as spatial wave-vector filters separating radiating from non-radiating velocity distributions. Cunefare [13] confirmed that the radiation efficiencies decay rapidly: only a small number of ARMs radiate to the far field, while higher-order modes contribute only to the evanescent near field. Elliott and Johnson [18] formalised this separation: in the sub-coincidence regime the radiation efficiency of mode k scales as $(kd/\lambda)^{2(k-1)}$, giving geometric decay with ratio $(d/\lambda)^2 < 1$. More recently, Ahrens and Sarradj [16] proposed a computationally efficient ARM calculation based on projecting spherical harmonics followed by a generalized SVD, avoiding the boundary integral equation formulation.

The correspondence is exact at the level of positive semi-definite quadratic forms and quotient metrics; membership in the φ-autosimilar admissibility class requires the additional spectral condition $\sigma_k \approx \varphi^{-k}$ over the relevant modal band. The radiation resistance matrix $\mathbf{R}$ plays the role of $K = P_5^* P_5$: it is a positive semi-definite quadratic form whose kernel contains all surface velocity distributions that radiate zero acoustic power to the far field. Two velocity distributions $\mathbf{v}_1$ and $\mathbf{v}_2$ satisfying $\mathbf{v}_1 - \mathbf{v}_2 \in \ker \mathbf{R}$ are *acoustically indistinguishable* in the far field regardless of their difference on the surface, precisely the statement $d_\partial([f],[g]) = 0$ if and only if $f - g \in \ker P_5$. The dictionary is:

$$\begin{aligned} \ker P_5 &\leftrightarrow \text{ evanescent velocity distributions (near field only)}, \\ \operatorname{Im} P_5 &\leftrightarrow \text{ propagating ARMs (far field)}, \\ d_\partial([f],[g]) = 0 &\leftrightarrow \text{ far-field indistinguishability}, \\ r = 5 &\leftrightarrow \text{ number of propagating ARMs above threshold } \sigma_{\text{cut}}. \end{aligned}$$

For a structure whose ARM singular spectrum satisfies $\sigma_k \approx \varphi^{-k}$ over the relevant modal band, the radiation resistance matrix falls within the admissibility class of Section 3. Theorem 4 then identifies $r = 5$ as the number of structurally admissible propagating ARMs: the bilateral threshold shows that retaining fewer than 5 modes excludes the transition cell w_4 that lies above the resolution threshold, while retaining more than 5 assigns positive metric weight to evanescent modes that produce identical far-field pressure, a physically incorrect discrimination.

A recurring practical issue in ARM-based active structural acoustic control is the choice of the control-basis rank [18, 13]: in practice engineers retain 3–5 modes empirically. Theorem 4 gives a structural criterion for the φ-autosimilar class without reference to geometry, frequency, or noise level. The rational filter $H(z) = P_4(z)/(1-\alpha z)$ is, in this context, the transfer function of the radiation filter mapping surface velocity measurements to far-field radiated power, retaining only the propagating ARM contribution and suppressing the evanescent kernel.

Remark 10. The acoustic interpretation applies only in the sub-coincidence regime, or in any modal band where the radiation spectrum satisfies $\sigma_k \approx \varphi^{-k}$. At and above coincidence the radiation efficiency spectrum can flatten, the transition threshold α can no longer separate propagating from evanescent behavior, and the admissibility class of Section 3 is not satisfied.

Consequences of Proposition 11 for ARM. The two properties of the Gram matrix $G = V_{\mathcal{N}}^* V_{\mathcal{N}}$ (Section B), together with the node-coherence diagnostic of Remark 12, have the following operator-theoretic interpretation in this setting.

Distinguishability (property i). The determinant $\det G = \prod_{i<j} |z_{n_i} - z_{n_j}|^2$ measures how well the five angular evaluation nodes can be separated in the retained representation. If two nodes were close on S^1, the corresponding ARMs would be nearly identical as surface velocity patterns, and no active control strategy could distinguish them regardless of sensor placement. The nodes $z_n = e^{i\pi/n}$ selected by the rank-5 constraint form a non-degenerate node set compatible with $\pi(n) \le r = 5$: $\det G$ is structurally non-zero, not by engineering design but because the prime-counting bound on admissible denominators forces the nodes apart.

Equidistribution and structural filter (property ii). The identity $G_{55}/\operatorname{tr} G = 1/5$ states that each of the five nodes z_n contributes equally to the overlap budget of $V_{\mathcal{N}}$: no node is

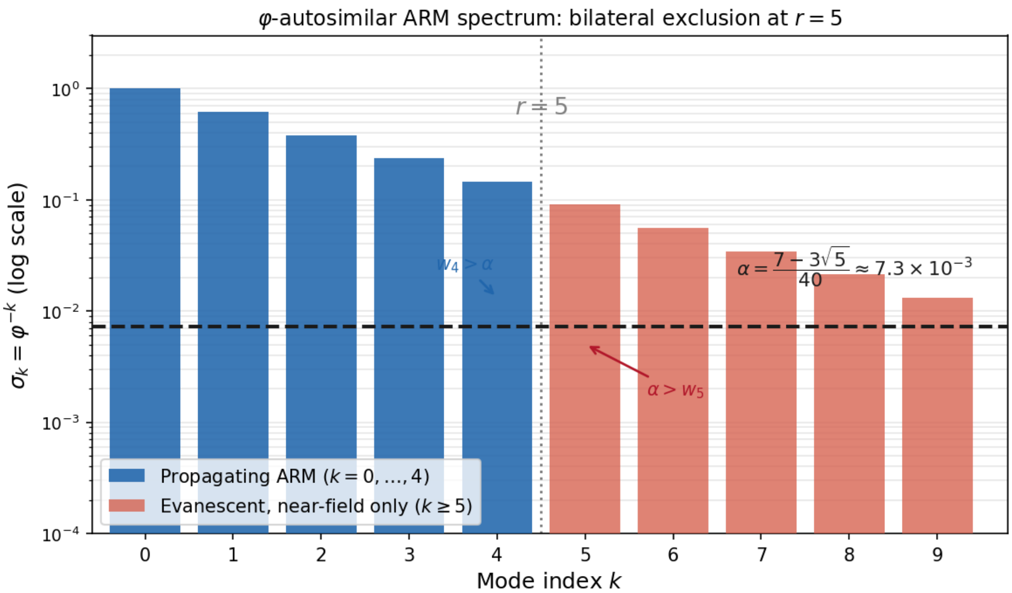

Figure 3: Mixed transition spectrum $w_k = \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)}$ for a Fibonacci-autosimilar radiation operator. Blue bars: resolvable transition cells $k = 0, \ldots, 4$, with $w_k > \alpha$. Red bars: unresolved transition cells $k \geq 5$, with $w_k < \alpha$. The structural threshold is $\alpha = \Lambda^{-2} \approx 7.3 \times 10^{-3}$. The bilateral exclusion of Theorem 4 gives the unique admissible rank $r = 5$ through $w_4 > \alpha > w_5$, without empirical fitting.

privileged. The modal energy shares are different — the dominant ARM ($k = 0$) carries $\approx 62\%$ of the radiated power and the fifth ($k = 4$) only $\approx 1.3\%$ — but the evaluation vectors $\mathbf{e}_n$ are equidistributed in self-overlap. The practical consequence is that the optimal structural projector $P_{\mathrm{opt}} = V_{\mathcal{N}} G^{-1} V_{\mathcal{N}}^*$ has unit rank at each node: the control system can reconstruct any radiated field in the admissibility class from five complex coefficients $\mathbf{c} = G^{-1} V_{\mathcal{N}}^* \mathbf{v}$, with equal numerical weight on each node. In active noise control this means: no node is a numerical bottleneck. The finite node separation $\min_{i \neq j} |z_{n_i} - z_{n_j}| \approx 0.163$ keeps $V_{\mathcal{N}}$ nonsingular; numerically $\mathrm{cond}(V_{\mathcal{N}}) \approx 2.25 \times 10^3$.

Node-coherence diagnostic. The node-coherence spectrum $C_k = |\sum_{n \in \mathcal{N}_{\mathrm{adm}}} z_n^k|^2$ (Remark 12) provides a structural envelope for the idealized admissibility class at each modal order. Numerically: $C_0 = 25$, $C_1 \approx 20.1$, $C_2 \approx 9.9$, $C_3 \approx 3.0$, $C_4 \approx 2.6$. These values depend only on $\mathcal{N}_{\mathrm{adm}}$, not on the geometry or frequency of a specific structure. They characterize the angular coherence of the retained nodes, not the radiated power of physical ARMs; actual modal energy shares are determined by the radiation resistance matrix and the structural velocity field.

5.2 Fibonacci photonic transfer spectra

A more rigid candidate realization of the φ-autosimilar admissibility class is suggested by one-dimensional Fibonacci photonic crystals, where the φ-autosimilar spectral scaling is generated by the geometry of the structure itself rather than imposed as a modelling assumption.

Construction. A one-dimensional Fibonacci photonic crystal is built from two dielectric layers A (refractive index n_A, thickness d_A) and B (n_B, d_B) stacked according to the Fibonacci substitution rule $S_{n+1} = S_n S_{n-1}$ with $S_1 = \mathrm{A}$, $S_2 = \mathrm{AB}$. The ratio of layer counts in successive generations satisfies $|S_{n+1}|/|S_n| \to \varphi$: the structure is φ-autosimilar by construction [9].

The optical transfer matrix of generation S_n satisfies a self-similar recursion

$$M_n = M_{n-1}M_{n-2}, \tag{43}$$

and Kohmoto, Kadanoff and Tang [9] showed that the trace of M_n obeys an invariant map whose dynamics determines the spectrum completely. Damanik and Gorodetski [17] proved that the allowed (pass) bands are dense but carry zero measure, while the forbidden (stop) bands (pseudo-gaps) carry all the spectral weight. The Fibonacci substitution forces a φ-autosimilar band hierarchy; in the idealized admissibility model this is represented by the scaling

$$\Delta_k \sim \varphi^{-k}. \tag{44}$$

The precise exponent and amplitude of the actual bandwidth depend on the optical contrast n_A/n_B, layer thicknesses, and Fibonacci generation; the φ^{-k} form is a structural model consistent with [10]. The mathematical structure of the Fibonacci spectrum — a Cantor set of zero Lebesgue measure with purely singular continuous spectral measures — has been established rigorously by Damanik and Gorodetski [17].

Propagating and evanescent modes. In finite Fibonacci stacks, pass-band windows support transmitted fields, while pseudo-gap windows support exponentially decaying fields inside the structure. In an input-output transfer formulation where singular values quantify transmitted pass-band amplitudes, the idealized Fibonacci-autosimilar regime is represented by a singular spectrum $\sigma_k \approx \varphi^{-k}$, provided this condition holds over the relevant modal band.

This places the Fibonacci photonic transfer operator within the admissibility class of Section 3: Theorem 4 then identifies $r = 5$ as the unique admissible rank of the idealized propagating transfer subspace. The separation is governed by the mixed transition weights $w_k = \sigma_k\sigma_{k+1} = \varphi^{-(2k+1)}$, not by comparing singular values σ_k directly with α. The admissible transition cells are those satisfying $w_k > \alpha$, giving the bilateral exclusion $w_4 > \alpha > w_5$. The evanescent sector corresponds to the kernel: it consists of internal field components that do not produce a distinguishable transmitted boundary response.

Figure 4 illustrates this for generation S_8 (34 layers, $n_A = 1.5$, $n_B = 2.3$, quarter-wave condition). The upper panel shows the transmittance spectrum computed by the transfer matrix method; shaded regions mark the primary pseudo-gaps. The lower panel shows the idealized mixed transition spectrum $w_k = \varphi^{-(2k+1)}$ with the structural threshold α and the bilateral exclusion boundary at $r = 5$.

Asymptotic scaling and structural protection. Unlike the structural acoustics case of the preceding section, where the φ-autosimilar spectrum is a property of a specific modal class, in the Fibonacci photonic crystal the φ^{-k} hierarchy emerges from the substitution rule $\{A \to AB,\ B \to A\}$ itself. In the infinite-generation limit the autosimilar scaling is exact [9, 10]; for finite generations the amplitude and exponent depend also on n_A, n_B, d_A, d_B. The admissibility class of Section 3 provides an operator-theoretic idealisation of this spectral structure.

The kernel sector contains modes that exist inside the crystal but produce no transmitted field at the boundary; Theorem 4 identifies the complementary propagating subspace as the unique admissible rank-5 projection.

The boundary metric d_∂ measures optical distinguishability between surface field distributions: two input field profiles f and g satisfying $f - g \in \ker P_5$ produce identical transmitted spectra, regardless of their difference at the input face.

Consequences of Proposition 11 for Fibonacci photonics. The two properties of G, together with the node-coherence diagnostic of Remark 12, acquire a spectral interpretation in the photonic setting.

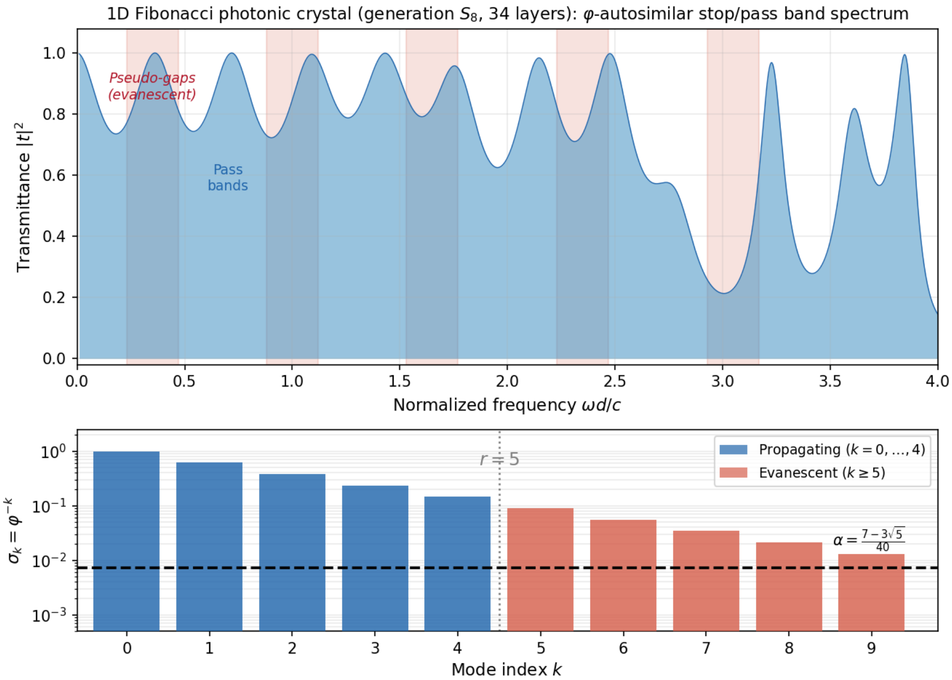

Figure 4: *Upper panel.* Transmittance $|t|^2$ of a 1D Fibonacci photonic crystal (generation S_8, 34 layers, $n_A = 1.5$, $n_B = 2.3$, quarter-wave condition), computed by the transfer matrix method. Shaded red regions mark primary pseudo-gaps; in these frequency windows the optical field decays exponentially inside the structure. Pass bands carry transmitted light in the finite stack. *Lower panel.* Idealized mixed transition spectrum $w_k = \sigma_k \sigma_{k+1} = \varphi^{-(2k+1)}$ associated with the Fibonacci-autosimilar admissibility model. Blue bars: resolvable transition cells $k = 0, \ldots, 4$, with $w_k > \alpha$. Red bars: unresolved transition cells $k \geq 5$, with $w_k < \alpha$. The structural threshold is $\alpha = \Lambda^{-2} \approx 7.3 \times 10^{-3}$. The bilateral exclusion of Theorem 4 gives the unique admissible rank $r = 5$ through $w_4 > \alpha > w_5$, without an empirical cut-off.

Band non-degeneracy (property i). The determinant $\det G = \prod_{i<j} |z_{n_i} - z_{n_j}|^2$ measures non-degeneracy of the five angular evaluation nodes in the idealized circular representative. A zero determinant would signal that two bands are degenerate — two retained node directions coincide in the idealized circular representative — which would mean two independently counted transmission channels are in fact the same channel. The nodes $z_n = e^{i\pi/n}$ with $n \in \mathcal{N}_{\mathrm{adm}}$ are non-degenerate by the prime-counting bound: $\det G \neq 0$ is a structural property of the admissibility class, not a consequence of specific layer parameters n_A, n_B, d_A, d_B.

Equidistribution of Bloch state overlap (property ii). The identity $G_{55}/\operatorname{tr} G = 1/5$ states that the five Bloch evaluation vectors $\mathbf{e}_n$ contribute equally to $\operatorname{tr} G$: each node z_n has the same self-overlap $G_{nn} = r = 5$. This is a statement about the *representational weight* of the nodes, not about the DOS. Its photonic content is: the transfer operator restricted to $\mathcal{N}_{\mathrm{adm}}$ is represented by a Gram matrix with equal diagonal — no node of the admissibility class is numerically dominant in the overlap basis. The practical consequence is that the structural projector $P_{\mathrm{opt}} = V_{\mathcal{N}} G^{-1} V_{\mathcal{N}}^*$ decomposes the transmitted field into five equally-weighted channels, and the finite node separation keeps $V_{\mathcal{N}}$ nonsingular; numerically $\operatorname{cond}(V_{\mathcal{N}}) \approx 2.25 \times 10^3$, independent of layer parameters n_A, n_B, d_A, d_B.

Node-coherence envelope. The node-coherence spectrum $C_k = |\sum_n z_n^k|^2$ (Remark 12), normalised to $C_0 = 25$, gives relative values $C_k/C_0 = \{1,\ 0.804,\ 0.397,\ 0.121,\ 0.104\}$ for $k = 0, 1, 2, 3, 4$. These are structurally fixed numbers for the node set $\mathcal{N}_{\mathrm{adm}}$ and provide a structural envelope for the idealized admissibility class. Physical transmittance depends additionally on layer parameters n_A, n_B, d_A, d_B and Fibonacci generation; the values C_k/C_0 are a node-coherence diagnostic, not a universal prediction of actual transmission for any specific realisation.

Conclusions

A single Fibonacci-autosimilar generator — the one-scale recursion $F_{N+1} = F_N + F_{N-1}$, with limit law $\varphi^2 = \varphi + 1$ — is not a one-dimensional object once it is made observable. Represented by a non-isometric compact projection at finite resolution, it has exactly five α-resolvable transition levels, and the induced rank-five projection P_5 defines three coupled structures: an observable quotient $\mathcal{H}/\ker P_5$ spanned by the five logarithmic Mellin jets, a boundary metric tensor $K = P_5^* P_5$ measuring distinguishability rather than geometric separation, and a Hilbert-like rational transfer representative $H(z) = P_4(z)/(1-\alpha z)$ encoding the same data scalar-wise on the autosimilar lattice. The scalar threshold α is the single quantity selecting all three: it gates the transition matrix $W = \Sigma S \Sigma$, it fixes the metric Gram matrix K, and it is the pole of $H(z)$.

Within the admissibility class of Section 3, a φ-autosimilar compact map $P : \mathcal{H} \to \mathcal{K}$ with singular values $\sigma_k = \varphi^{-k}$ has a unique admissible SVD truncation rank, equal to the cardinality of the α-resolvable transition spectrum, $r = \#\{k : \sigma_k \sigma_{k+1} \geq \alpha\} = 5$ (Theorem 4). Both bounds follow from the position of the single threshold α within the transition weights: the lower bound $r \geq 5$ from $w_4 > \alpha$ (level 4 must be retained) and the upper bound $r \leq 5$ from $\alpha > w_5$ (level 5 must be excluded), with α derived independently from the Pincherle operator and Fibonacci curvature (Remark 1). The selected rank coincides with the Fibonacci closure capacity $W = |\lambda_+ - \lambda_-|^2 = 5$, permitting the five-sector phasorial realization of Corollary 5; that agreement is a consequence of the result, not an input to it.

Once $r = 5$ is established, the remaining structures are determined by the same finite-rank operator without additional assumptions:

- the observable basis $V_5 = \operatorname{span}\{(\log x)^n\}_{n=0}^4$ (retained column space of $P^{(5)}$);
- the boundary metric $K = P_5^* P_5$ on the quotient $\mathcal{H}/\ker P_5$ (induced quadratic form);
- the rational transfer representative $H(z) = P_4(z)/(1-\alpha z)$ (scalar representative of K on the autosimilar lattice).

These three are not independent outputs: they are the column space, the Gram form, and the scalar transfer function of one rank-five operator.

Moore–Penrose interface. Because P_5 is non-invertible ($\ker P_5 \neq \{0\}$), the canonical reconstruction from a boundary observation $u \in \operatorname{Im} P_5$ is the Moore–Penrose pseudoinverse

$$P_5^+ : \operatorname{Im} P_5 \longrightarrow (\ker P_5)^\perp, \qquad P_5^+ u = \arg\min\{\|x\|_{\mathcal{H}} : P_5 x = u\}. \tag{45}$$

The pseudoinverse selects the minimum-norm bulk preimage of each boundary observation, landing in $(\ker P_5)^\perp$, the metrically visible subspace. The metric $K = P_5^* P_5$ governs which distinctions survive the projection; P_5^+ governs what minimum-norm bulk state corresponds to each surviving distinction. Together, K and P_5^+ are the two faces of the same finite-rank structure: K reads the boundary, P_5^+ reconstructs the minimum bulk consistent with that reading.

An optional geometric readout is developed in Appendix B: the rank-five structure admits a finite catalog of admissible angular denominators $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ on S^1, connected to the same Vandermonde and Gram-matrix objects established by the main theorem.

All results are conditional on Requirements (I)–(III).

Declarations. The author declares no competing interests and received no funding. All results are analytic; figures are generated from the formulas in the text.

A Numerical implementation of $H(z)$ and rank selection

The rational filter $H(z) = P_4(z)/(1-\alpha z)$ is implemented as a standard IIR filter with numerator $B(z^{-1})$ and denominator $A(z^{-1})$:

$$B = [b_0, b_1, b_2, b_3, b_4], \quad b_n = (\log\varphi)^n, \qquad A = [1, -\alpha], \tag{46}$$

with $\alpha = \Lambda^{-2} \approx 7.295 \times 10^{-3}$ and $\log\varphi \approx 0.4812$. The pole at $z = \alpha^{-1} = (F_5 + F_4\sqrt{5})^2 = 70 + 30\sqrt{5} \approx 137$ lies outside the unit disk. The causal DSP implementation uses

$$H_c(z) = \frac{P_4(z^{-1})}{1 - \alpha z^{-1}}, \tag{47}$$

whose pole is at $z = \alpha$, strictly inside the unit disk, so H_c is stable and proper.

On the autosimilar lattice $\{z_k = \varphi^{-k}\}_{k=0}^{9}$ the filter evaluates to

$$H(z_k) = \frac{p^T z_k}{1 - \alpha z_k}, \qquad p = [1, \log\varphi, (\log\varphi)^2, (\log\varphi)^3, (\log\varphi)^4]^T, \qquad \mathbf{z}_k = [1, z_k, z_k^2, z_k^3, z_k^4]^T, \tag{48}$$

with values $H(z_0) \approx 1.892$, $H(z_1) \approx 1.426$, $H(z_2) \approx 1.228$, $H(z_3) \approx 1.130$, $H(z_4) \approx 1.077$, all decreasing monotonically, consistent with the spectral weighting $\sigma_k^2 = \varphi^{-2k}$.

Figure 5 compares the structural rank-selection criterion of Theorem 4 with three standard empirical methods: energy thresholds at 90%, 99%, and 99.9% of cumulative spectral energy. (The spectral gap criterion is not applicable here: since $\sigma_k = \varphi^{-k}$, all log-gaps $\log(\sigma_k/\sigma_{k+1}) = \log\varphi$ are equal, so no gap is larger than any other.) The spectral weights $w_k = \varphi^{-(2k+1)}$ straddle the threshold α between $k = 4$ and $k = 5$ (panel A); the filter $H(z_k)$ and the metric weights σ_k^2 are both monotone decreasing with a natural cutoff at $k = 5$ (panel B). The cumulative energy (panel C) shows that the 99% threshold selects $r = 5$, while the 90% threshold selects $r = 3$ and the 99.9% threshold selects $r = 8$: the empirical choice is unstable with respect to the threshold level. The structural method (panel D) selects $r = 5$ without any free parameter.

The Python implementation uses `numpy` and `scipy.signal`:

```
import numpy as np
from scipy import signal

phi    = (1 + np.sqrt(5)) / 2
Lambda = 2 * phi**2 * np.sqrt(5)        # Pincherle increment x curvature
alpha  = Lambda**-2                     # = 1/(20*phi**4) = (7-3*sqrt5)/40
u      = np.log(phi)

b = np.array([u**n for n in range(5)])  # numerator B
a = np.array([1.0, -alpha])             # denominator A

# Apply to signal x:
y = signal.lfilter(b, a, x)

# Evaluate on autosimilar lattice:
k   = np.arange(5)
z_k = phi**(-k)
p   = np.array([u**n for n in range(5)])
H_k = np.array([p @ [zk**n for n in range(5)]
                 / (1 - alpha*zk) for zk in z_k])
```

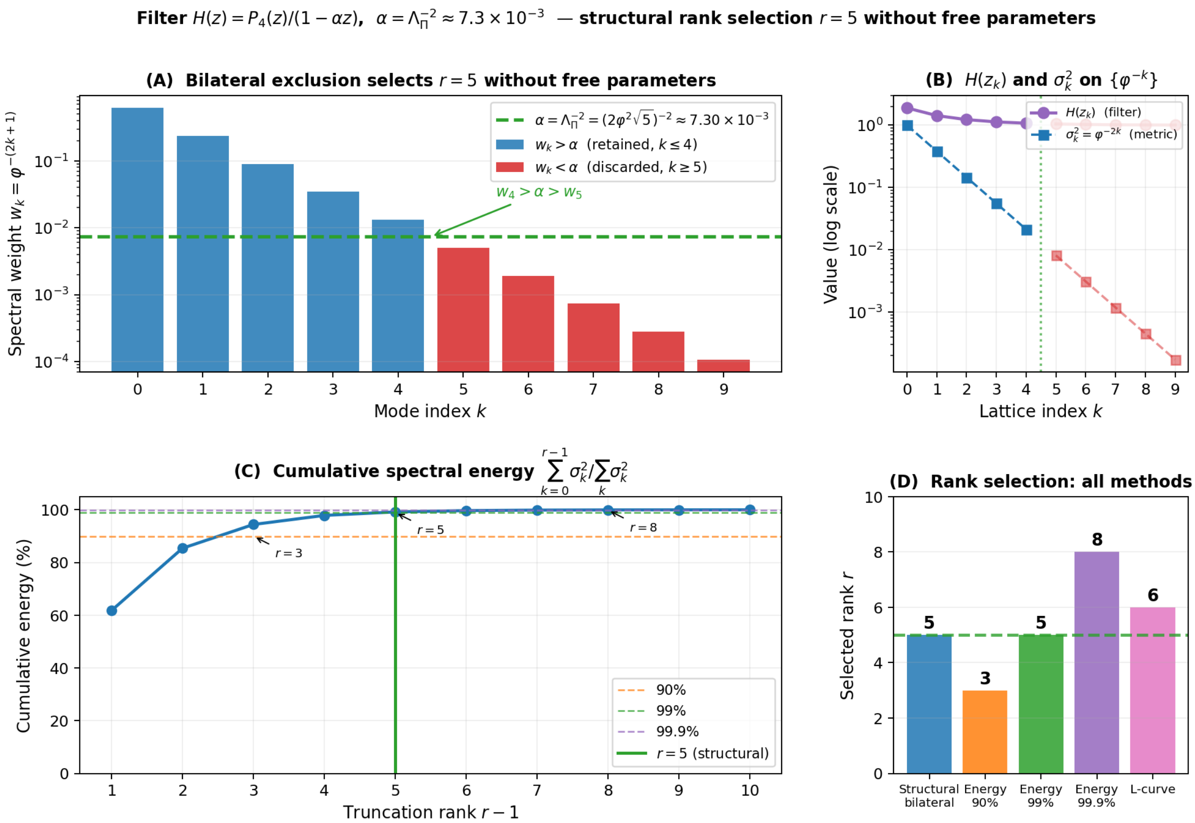

Figure 5: Numerical verification of the rank-selection criterion. **(A)** Transition weights $w_k = \varphi^{-(2k+1)}$ on logarithmic scale; dashed line: threshold $\alpha = \Lambda^{-2} \approx 7.3 \times 10^{-3}$. The bilateral exclusion $w_4 > \alpha > w_5$ is exact. **(B)** Filter values $H(z_k)$ and metric weights σ_k^2 on the autosimilar lattice $\{z_k = \varphi^{-k}\}$; both decay monotonically with a natural cutoff at $k = 5$. **(C)** Cumulative spectral energy: the 99% level coincides with $r = 5$, while 90% selects $r = 3$ and 99.9% selects $r = 8$. (The spectral gap criterion is not applicable here: all log-gaps equal $\log \varphi$, so no gap is larger than any other.) **(D)** Summary: the structural bilateral exclusion and the 99% energy threshold both select $r = 5$; the empirical choice is unstable with respect to the threshold level, while the structural method requires no free parameter.

B Geometric Corollary: Holonomy and Angular Residual (optional readout)

Theorem 4 fixes $r = 5$ and Sections 4 establishes V_5, K, and $H(z)$. A further consequence of the rank-five structure is a finite catalog of admissible angular denominators on S^1, obtained by asking which primitive roots of unity $z_n = e^{i\pi/n}$ close within the retained orbit. This section derives that catalog using the circular Vandermonde matrix $V_{\mathcal{N}}$ with nodes $\{z_n\}_{n\in\mathcal{N}_{\mathrm{adm}}}$; the Gram matrix $G = V_{\mathcal{N}}^* V_{\mathcal{N}}$ connects the angular readout to the same linear-algebraic objects established in Sections 3–4.

This section derives a geometric consequence of the rank-five characterization and is not required for the main rank-selection theorem. It may be read independently.

Circular Vandermonde matrix. Associate to each $n \in \mathcal{N}_{\mathrm{adm}}$ the evaluation vector

$$\mathbf{e}_n \;:=\; \big[1,\; z_n,\; z_n^2,\; z_n^3,\; z_n^4\big]^\top \;\in\; \mathbb{C}^r, \qquad z_n = e^{i\pi/n}, \tag{49}$$

where $z_n = e^{i\pi/n}$ is the primitive $2n$-th root of unity. Form the *circular Vandermonde matrix*

$$V_{\mathcal{N}} \;:=\; \big[\,\mathbf{e}_2 \,\big|\, \mathbf{e}_3 \,\big|\, \mathbf{e}_5 \,\big|\, \mathbf{e}_7 \,\big|\, \mathbf{e}_{11}\,\big] \;\in\; \mathbb{C}^{r\times|\mathcal{N}_{\mathrm{adm}}|}. \tag{50}$$

This matrix encodes the angular sampling points on S^1 associated with the admissible nodes: its columns are evaluation vectors at the five primitive roots of unity forced by the rank-five constraint. Vandermonde matrices with nodes on the unit circle have been studied in [14, 15]; the present matrix belongs to this class with nodes $z_n \in S^1$. The Gram matrix of $V_{\mathcal{N}}$ has entries

$$G_{ij} \;:=\; \mathbf{e}_{n_i}^*\,\mathbf{e}_{n_j} \;=\; \sum_{k=0}^{r-1} \overline{z_{n_i}^k}\, z_{n_j}^k \;=\; \sum_{k=0}^{r-1}\left(\frac{z_{n_j}}{z_{n_i}}\right)^k \;=\; \frac{1-\big(z_{n_j}/z_{n_i}\big)^r}{1-z_{n_j}/z_{n_i}}, \qquad i \neq j, \tag{51}$$

and $G_{ii} = \|\mathbf{e}_{n_i}\|^2 = r = 5$. (Here $\mathbf{e}^*\mathbf{f} = \sum_k \overline{e_k} f_k$ is the standard Hermitian inner product; $G = V_{\mathcal{N}}^* V_{\mathcal{N}}$ with $V_{\mathcal{N}}[k,i] = z_{n_i}^k$.) Since $|z_{n_i}| = |z_{n_j}| = 1$ and $z_{n_i} \neq z_{n_j}$ for $n_i \neq n_j \in \mathcal{N}_{\mathrm{adm}}$, the matrix G is non-singular: $V_{\mathcal{N}}$ has rank $|\mathcal{N}_{\mathrm{adm}}| = 5 = r$.

The entry G_{ij} is a geometric partial sum of the unweighted reproducing kernel of the retained boundary subspace at the node pair (z_{n_i}, z_{n_j}):

$$G_{ij} \;=\; \sum_{k=0}^{r-1} e^{ik\pi(1/n_j - 1/n_i)} \;=\; \Pi_5\big|_{(z_{n_i}, z_{n_j})}, \tag{52}$$

where $\Pi_5 = U_5 U_5^*$ is the unweighted orthogonal projector onto $\mathrm{Range}(P_5)$. Note: $P_5 P_5^* = U_5 \Sigma_5^2 U_5^*$ has weights $\sigma_k^2 = \varphi^{-2k}$ and is *not* equal to G; the weighted version

$$\widetilde{G}_{ij} \;=\; \sum_{k=0}^{r-1} \sigma_k^2\, e^{ik\pi(1/n_j - 1/n_i)} \;=\; (P_5 P_5^*)\big|_{(z_{n_i}, z_{n_j})}$$

gives the restriction of $P_5 P_5^*$ to the same nodes and is the metric kernel K restricted to the node pairs. The diagonal $G_{ii} = r = 5$ and the off-diagonal formula (51) connect the SVD rank to the adjacency structure: G_{ii} counts the r retained singular modes, while $\mathrm{spec}(A_{n_i}) = \{2\cos(k\pi/n_i) : k = 0, \ldots, 2n_i - 1\}$ are the eigenvalues of the cyclic graph $G_{n_i} = C_{2n_i}$ generated by the orbit of z_{n_i}.

Spectral properties of G. Two properties of the Gram matrix $G = V_{\mathcal{N}}^* V_{\mathcal{N}}$ follow directly from the Vandermonde structure and are stated here as a proposition.

Proposition 11 (Spectral properties of the circular Gram matrix)**.** *Let* $G = V_{\mathcal{N}}^{*} V_{\mathcal{N}}$ *with* $V_{\mathcal{N}} \in \mathbb{C}^{r\times|\mathcal{N}_{\mathrm{adm}}|}$ *as in* (50)*. Then:*

(i) **Vandermonde determinant.**

$$\det G \;=\; \prod_{\substack{i<j \\ n_i,n_j\in\mathcal{N}_{\mathrm{adm}}}} |z_{n_i} - z_{n_j}|^2.$$

(ii) **Trace partition.**

$$\operatorname{tr} G = r\,|\mathcal{N}_{\mathrm{adm}}| = 25, \qquad G_{55} = r = 5, \qquad \frac{G_{55}}{\operatorname{tr} G} = \frac{1}{|\mathcal{N}_{\mathrm{adm}}|}.$$

The lock-in node $n = 5$ *contributes exactly* $1/|\mathcal{N}_{\mathrm{adm}}|$ *of the total trace: its self-overlap equals the mean self-overlap of all nodes.*

Proof. (i) follows from $\det(V^*V) = |\det V|^2$ (Cauchy–Binet) and the Vandermonde identity $\det V_{\mathcal{N}} = \prod_{i<j}(z_{n_i} - z_{n_j})$.

(ii) $\operatorname{tr} G = \sum_n \|\mathbf{e}_n\|^2 = r\,|\mathcal{N}_{\mathrm{adm}}|$ since $\|\mathbf{e}_n\|^2 = r$ for every n (each z_n lies on S^1). $G_{55} = \|\mathbf{e}_5\|^2 = r$. The ratio follows. □

Remark 12 (Node-coherence spectrum)**.** Independently of the Gram matrix G, the admissible nodes define the *node-coherence spectrum*

$$C_k \;:=\; \Big|\sum_{n\in\mathcal{N}_{\mathrm{adm}}} z_n^k\Big|^2, \qquad k = 0,\ldots,r-1. \tag{53}$$

This quantity measures the coherent phasing of the retained angular nodes at the k-th autosimilar orbit frequency. It is a diagnostic of the node set $\mathcal{N}_{\mathrm{adm}}$, not a property of $G = V_{\mathcal{N}}^{*} V_{\mathcal{N}}$. Numerically, $C_0 = 25$, $C_1 \approx 20.09$, $C_2 \approx 9.93$, $C_3 \approx 3.01$, $C_4 \approx 2.59$. The identity $C_0 = \operatorname{tr} G$ reflects that all nodes lie on S^1 and that $|\mathcal{N}_{\mathrm{adm}}| = r$.

Remark 13 (Meaning of property (ii))**.** Property (ii) does not say that $n = 5$ carries one fifth of the *energy* of the singular modes: the modal energy shares are $\sigma_k^2/\sum_j \sigma_j^2 = \varphi^{-2k}/\sum_j \varphi^{-2j}$, which for $k = 0,\ldots,4$ give approximately 62%, 24%, 9%, 3%, 1%. What property (ii) says is that all five nodes z_n have identical self-overlap $G_{ii} = r = 5$: they contribute equally to $\operatorname{tr} G = r|\mathcal{N}_{\mathrm{adm}}|$. The identity $G_{55}/\operatorname{tr} G = 1/|\mathcal{N}_{\mathrm{adm}}|$ is therefore a statement of *equidistribution* of evaluation weight across nodes, not of modal energy. Its consequence for signal processing is concrete: any φ-autosimilar signal can be represented in the basis $V_{\mathcal{N}}$ by five complex coefficients $\mathbf{c} = G^{-1}V_{\mathcal{N}}^{*}\mathbf{y}$, and each node contributes equally to the representation cost — the optimal projector $P_{\mathrm{opt}} = V_{\mathcal{N}} G^{-1} V_{\mathcal{N}}^{*}$ is idempotent with rank $r = 5$ and has no preferred node.

Closure residual and admissible nodes.

The retained singular subspace has dimension $r = 5$, with independent samples indexed by $k = 0,\ldots,r-1$. The angular orbit closes at the endpoint $k = r$, which is not an additional singular direction but the closing point of the retained cycle. The angular readout set is therefore

$$\Theta_r \;:=\; \big\{e^{ik\pi/5} : k = 0,\ldots,r\big\}, \tag{54}$$

with $k = r$ understood as the closure endpoint ($e^{ir\pi/5} = e^{i\pi} = -1$, the antipodal point of the orbit, not an additional retained mode).

For a node $z_n = e^{i\pi/n}$, closure on the principal half-cycle requires $z_n^j = e^{ik\pi/5}$ for some $1 \le j \le r$ and $0 \le k \le r$. The angular closure residual is

$$\delta_n \;:=\; \min_{\substack{1\le j\le r\\ 0\le k\le r}} \left| \frac{j}{n} - \frac{k}{5} \right|. \tag{55}$$

Closure nodes are precisely those with $\delta_n = 0$; for nodes with $\delta_n > 0$, the angular information is present in the circular Vandermonde catalog but the orbit does not close within the retained rank-five cycle. The condition $\delta_n = 0$ is equivalent to the divisibility $5j = kn$ for some (j, k) in the admissible range, and the minimal such j is

$$j_{\min}(n) \;=\; \frac{n}{\gcd(5, n)}. \tag{56}$$

The node z_n is a closure node if and only if $j_{\min}(n) \le r$.

Admissible angular catalog.

Corollary 14 (Admissible angular catalog)**.** *The admissible set of irreducible angular denominators induced by the rank-5 bound $\pi(n) \le r = 5$ is*

$$\mathcal{N}_{\mathrm{adm}} = \{2,\ 3,\ 5,\ 7,\ 11\}. \tag{57}$$

The cutoff at $n = 11$ is the saturation point: $\pi(11) = 5$ and $\pi(13) = 6 > 5$.

Proof. The admissibility class is restricted to prime denominators n by definition: a composite $n = ab$ factors through coarser angular partitions already represented by z_a and z_b, contributing no new angular resolution to the φ-autosimilar orbit. This is a class restriction, not a linear-independence claim about Vandermonde columns (which are all independent when nodes are distinct). Within this class, the count of primes $\pi(n)$ is the angular dimension; the bound $\pi(n) \le r = 5$ and the saturation $\pi(11) = 5$, $\pi(13) = 6 > r$ force $n = 11$ as the largest admissible denominator and $n = 13$ as the first excluded one. □

Among the five elements of $\mathcal{N}_{\mathrm{adm}}$, the closure criterion $j_{\min}(n) \le r$ selects the *closure nodes.* Using $j_{\min}(n) = n/\gcd(5, n)$:

$$\begin{array}{lll}
n = 5: & j_{\min} = 5/5 = 1 \le 5. & \text{Closure node. } \checkmark\ (2\cos(\pi/5) = \varphi) \\
n = 2: & j_{\min} = 2/1 = 2 \le 5. & \text{Closure node. } \checkmark \\
n = 3: & j_{\min} = 3/1 = 3 \le 5. & \text{Closure node. } \checkmark \\
n = 7: & j_{\min} = 7/1 = 7 > 5. & \text{Not a closure node.} \\
n = 11: & j_{\min} = 11/1 = 11 > 5. & \text{Saturation node only.}
\end{array} \tag{58}$$

The vector $\mathbf{e}_7$ is a column of $V_{\mathcal{N}}$ and contributes to the full Vandermonde rank $r = 5$. However, z_7 is not a closure node: $\delta_7 > 0$, so the orbit $\{z_7^j\}$ does not meet Θ_r within $j \le r$ steps.

Scale multiplier and boundary catalog. The scale multiplier between adjacent levels is

$$S = n_{\min} \cdot r = 2 \times 5 = 10, \tag{59}$$

giving boundary scales $\{\alpha \cdot 10^k\}_{k=0}^4$. The map $k(n) = r - \pi(n)$ assigns each closure node and the saturation point to a scale level. The complete correspondence is in Table 1.

Remark 15 ($n = 7$ in $\mathcal{N}_{\mathrm{adm}}$ but not a closure node)**.** The element $n = 7$ satisfies $\pi(7) = 4 \le r = 5$, so its column $\mathbf{e}_7$ is included in $V_{\mathcal{N}}$ and contributes to its rank. However, $j_{\min}(7) = 7/\gcd(5, 7) = 7 > r = 5$, so $\delta_7 > 0$ and the orbit of z_7 does not meet Θ_r within r steps. Thus $n = 7$ is admissible as an angular denominator but does not define a closure node of the retained rank-five orbit.

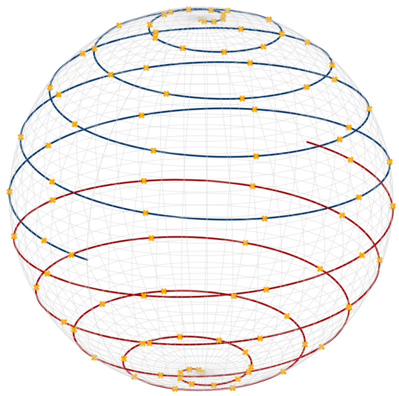

Figure 6: Optional geometric visualization of the φ-autosimilar scale hierarchy as windings on S^2: successive scale levels $x_k = \varphi^k$ are mapped to a constant-azimuthal-increment orbit, sampled at the lattice points (orange). This picture is a pre-projection visualization of bulk scale structure; the rank characterization of Theorem 4 is fixed by the threshold position $w_4 > \alpha > w_5$ and does *not* use this construction or any $S^2 \to S^1$ reduction. The boundary readout is obtained only after projection, as the discrete angular catalog below.

Remark 16 (Number-theoretic stratification of the catalog)**.** The admissible catalog $\mathcal{N}_{\text{adm}} = \{2, 3, 5, 7, 11\}$ carries a quadratic-field stratification, recorded here as a property of the catalog and *not used* in the rank characterization of Theorem 4. The node $n = 5$ is the real quadratic generator: $\mathbb{Q}(\sqrt{5})$ has discriminant $5 \equiv 1 \pmod 4$ and ring of integers $\mathcal{O} = \mathbb{Z}[(1+\sqrt{5})/2] = \mathbb{Z}[\varphi]$, generated by the autosimilar unit φ. Among the remaining denominators, $n = 3, 7, 11$ are Heegner numbers whose imaginary quadratic fields $\mathbb{Q}(\sqrt{-n})$ share the same half-integral ring of integers,

$$\mathcal{O}_{\mathbb{Q}(\sqrt{-n})} = \mathbb{Z}\left[\tfrac{1+\sqrt{-n}}{2}\right], \qquad -n \equiv 1 \pmod 4 \quad (n = 3, 7, 11);$$

the node $n = 2$ is the exceptional Heegner case $-2 \equiv 2 \pmod 4$, with the pure ring $\mathcal{O}_{\mathbb{Q}(\sqrt{-2})} = \mathbb{Z}[\sqrt{-2}]$. The stratification is therefore 5 (real, half-integral) / 3, 7, 11 (imaginary Heegner, half-integral) / 2 (imaginary Heegner, pure-integral). The minimal node $n = 2$ also fixes the factor 2 of the boundary-scale step $10 = n_{\min} \cdot r = 2 \cdot 5$ used in Table 1. The convention $\sqrt{-n}$ denotes the principal branch; the half-integral generator is written $(1 + \sqrt{-n})/2$, never with an explicit factor i.

Table 1 summarises the boundary scale structure derived from $V_{\mathcal{N}}$. The node z_7 is not a closure node: $j_{\min}(7) = 7 > r = 5$, so $\delta_7 > 0$ and the orbit $\{z_7^j\}$ does not meet Θ_r within r steps. The three closure nodes z_n ($n \in \{2, 3, 5\}$), together with the saturation node $n = 11$, generate the complete observable boundary algebra from $\mathcal{N}_{\text{adm}}$, with no free parameter.

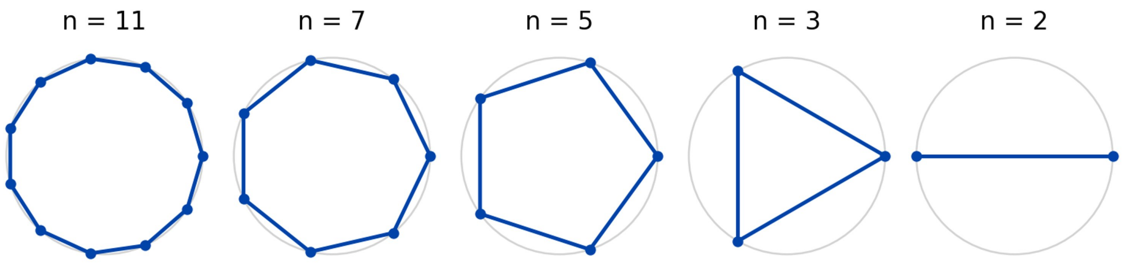

Figure 7: Cyclic graphs $G_n = C_{2n}$ generated by the orbits $\{z_n^k : k = 0, \ldots, 2n-1\}$ of the primitive $2n$-th roots of unity $z_n = e^{i\pi/n} \in S^1 \subset \mathbb{C}$, for $n \in \mathcal{N}_{\mathrm{adm}} = \{2, 3, 5, 7, 11\}$ and the fine-grid limit $n \gg 1$. Each graph is a discrete S^1-readout associated with one admissible angular denominator. The adjacency matrix A_n of G_n has eigenvalues $\lambda_k = 2\cos(k\pi/n)$; among all G_n with $n \in \mathcal{N}_{\mathrm{adm}}$, only $G_5 = C_{10}$ satisfies $\varphi \in \mathrm{spec}(A_5)$, i.e. $\varphi = 2\cos(\pi/5)$. The dashed arc illustrates a non-zero angular closure residual for a non-closing node ($n = 7$).

Table 1: Closure nodes and saturation point from $\mathcal{N}_{\mathrm{adm}}$, with boundary scales $\alpha \cdot 10^{k(n)}$ derived from $j_{\min}(n) = n/\gcd(5, n)$ and $k(n) = r - \pi(n)$. $n = 7$ is not a closure node ($j_{\min} = 7 > r$) and has no boundary scale entry.

n	$z_n = e^{i\pi/n}$	$j_{\min}(n)$	$\pi(n)$	$k(n)$	Boundary scale
11	$e^{i\pi/11}$	11 (sat.)	5	0	α
5	$e^{i\pi/5}$	1	3	2	$\alpha \cdot 10^2$
3	$e^{i\pi/3}$	3	2	3	$\alpha \cdot 10^3$
2	$e^{i\pi/2}$	2	1	4	$\alpha \cdot 10^4$

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